



Politecnico di Torino

Qubit Electronics

Lab 6

A. Tudisco, G. Piccinini, F. Riente

27 maggio 2025

1 Introduction

During this laboratory session, we will focus on the energy level splitting in the SiMOS double quantum dot device. It is recommended to run the Python code at home using the server. Here are some useful commands:

- `tmux new -s nameSession`
to create a new background terminal session called **nameSession**.
- `tmux attach -t nameSession`
to attach to the background terminal session **nameSession**.
- `tmux kill-session -t nameSession`
to kill the background terminal session **nameSession**.

To detach from the background terminal session, use:

`ctrl + b + d`

2 Python script

Given the provided Python code, add the Schrodinger and Poisson solvers inside the for loop, where indicated. Then, add the required lines of code to calculate the energy difference between the two energy levels (ground state and first excited). Save the results related to the difference and the inter-dot barrier vector.

Question: Which parameters related to the structure differ from the previous laboratory?

Question: Report the line of code added.

3 Analysis of the results

Write a script to perform an exponential fit of the values of the energy level splitting versus the inter-dot barrier and comment on the results obtained. Is the behavior for low inter-dot barrier gate bias voltage as expected? Explain why or why not.